

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division

2070 Chaitra

Exam	Regular/Back		
Level	M.Sc.	Full Marks	60
Programme	Computer system and Knowledge Engineering	Pass Marks	30
Year/Part	I/II	Time	3 hrs.

Subject- Image Processing (Elective)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figure in the margin indicate **Full Marks**
- ✓ Assume suitable data if necessary.

- (1) Give the definition of digital image and draw only the block diagram of a typically digital image processing system? How many images of size 1200×800 with 8 bit gray value can be stored in 64 MB storage space ? [1+2+4=7]
- (2) What is the Discrete cosine transform (DCT) ? Explain how can you apply it in digital image processing along with the properties of DCT. [3+2+2=7]
- (3) Explain in brief the term “contrast stretching” using histogram in digital image Processing ? Explain the adaptive median filter with suitable algorithm for its implementation. [4+4=8]
- (4) What is zooming ? A 3×3 size image has following intensity Information. [1+5=6]

1	5	3
6	4	2
1	7	6

Change this image into 66 size image using zooming by interpolation method.
- (5) Given the following frequency table obtained from histogram of 16×16 , 8 level image. [6+1+1=8]

M	0	1	2	3	4	5	6	7
N_M	6	15	16	70	35	31	51	32

Where M = gray level and N_M = pixels having Mth gray level.
Construct Huffman code for each gray level. Also calculate the compression ratio and coding redundancy for the system.
- (6) Derive the equation for elliptical filter and write the algorithm for its Implementation. [4+3=7]
- (7) What do you mean by segmentation? Explain how do you select the threshold in segmentation using any one techniques. [1+7=8]
- (8) Write short notes on: [3×3=9]
 - a) Butterworth High Pass Frequency Domain Filter.
 - b) Lossy Predictive Coding
 - c) Opening and closing morphological operations

